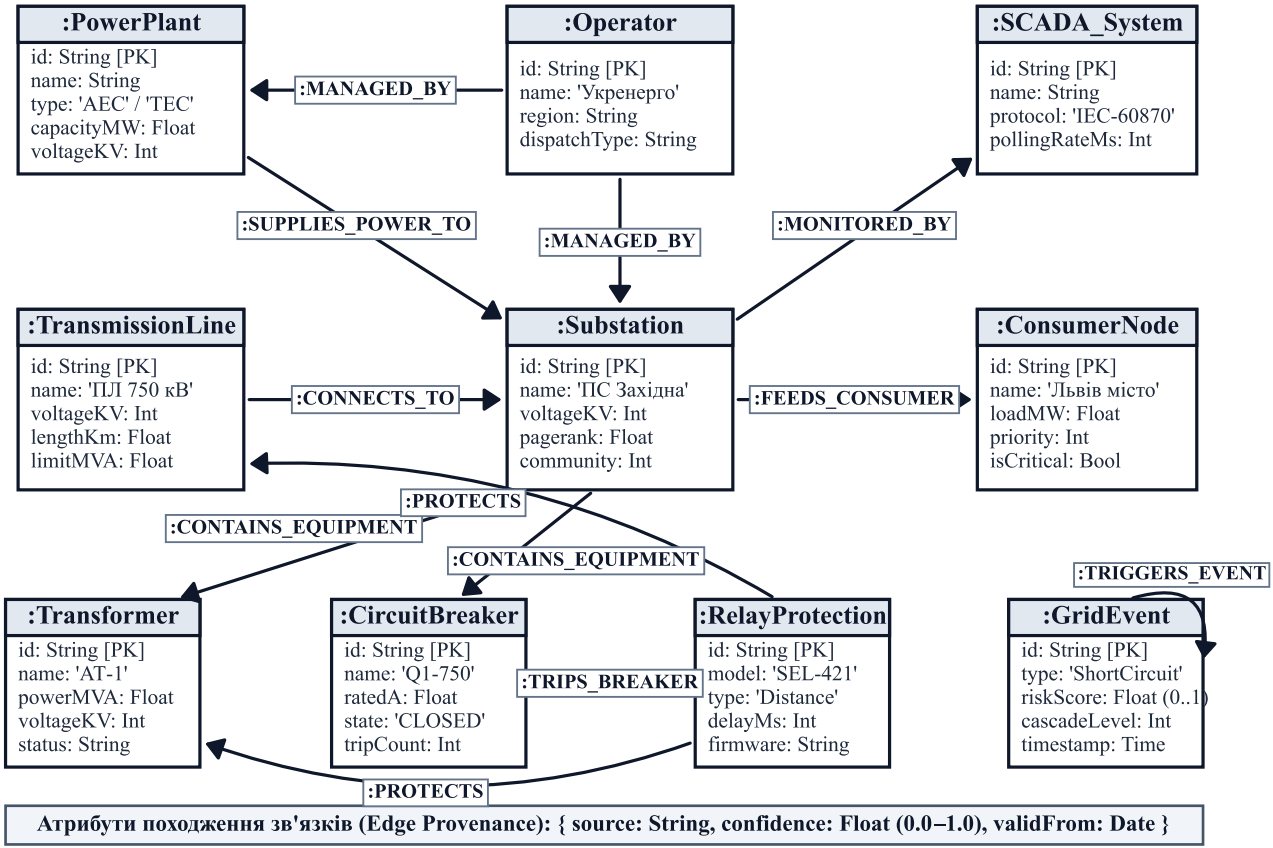

:PowerPlant
id: String [PK]
name: String
type: 'AEC' / 'TEC'
capacityMW: Float
voltageKV: Int
:Operator
id: String [PK]
name: 'Укренерго'
region: String
dispatchType: String
:SCADA_System
id: String [PK]
name: String
protocol: 'IEC-60870'
pollingRateMs: Int
:MANAGED_BY
:SUPPLIES_POWER_TO
:MANAGED_BY
:MONITORED_BY
:TransmissionLine
id: String [PK]
name: 'ПЛ 750 кВ'
voltageKV: Int
lengthKm: Float
limitMVA: Float
:Substation
id: String [PK]
name: 'ПС Західна'
voltageKV: Int
pagerank: Float
community: Int
:ConsumerNode
id: String [PK]
name: 'Львів місто'
loadMW: Float
priority: Int
isCritical: Bool
:CONNECTS_TO
:FEEDS_CONSUMER
:PROTECTS
:CONTAINS_EQUIPMENT
:CONTAINS_EQUIPMENT
:TRIGGERS_EVENT
:Transformer
id: String [PK]
name: 'AT-1'
powerMVA: Float
voltageKV: Int
status: String
:CircuitBreaker
id: String [PK]
name: 'Q1-750'
ratedA: Float
state: 'CLOSED'
tripCount: Int
:TRIPS_BREAKER
:RelayProtection
id: String [PK]
model: 'SEL-421'
type: 'Distance'
delayMs: Int
firmware: String
:GridEvent
id: String [PK]
type: 'ShortCircuit'
riskScore: Float (0..1)
cascadeLevel: Int
timestamp: Time
:PROTECTS
Атрибути походження зв'язків (Edge Provenance): { source: String, confidence: Float (0.0–1.0), validFrom: Date }